System Design

System Design

how a software system should be structured, how its components communicate, how data flows through it, and how it will handle real-world requirements such as performance, security, reliability, scalability, and maintainability.

System design is the blueprint of a software system before and during implementation.

Frontend → Backend → Database

System Design = architecture + components + communication + data + operational behavior

Types of System Design
High-Level Design — HLD (**overall architecture**.)
Low-Level Design — LLD (describes **how individual components are implemented internally**.)

Functional Requirements - Before designing a system, identify what the system must do.
Non-Functional Requirements - describe how well the system should operate.

Stateless Architecture - A server is stateless when it doesn't depend on local memory to maintain user session state.

context diagram - shows the **system as a single high-level entity** and the external entities that exchange information with it.

Know what is happening → know what will be affected → know what action can be taken

Core Problem

> The core problem is the lack of connected, timely, and actionable visibility across procurement, suppliers, inventory, and production, making it difficult for manufacturing organizations to identify supply risks early and take informed corrective action.

Proposed Product

An Intelligent Supply Chain Visibility and Procurement Decision-Support Platform for Smart Manufacturing.

Vision

Help manufacturing organizations detect supply-chain risks early, understand their business impact, and make faster, data-driven procurement decisions.

Supplier Management
Procurement Management
RFQ & Supplier Quotation
Supplier Evaluation
Purchase Order Management
Supply Chain Visibility Dashboard
Supplier Risk Monitoring
Risk Score
Inventory & Material Visibility
Production Impact Analysis
Alerts & Notifications
AI-Assisted Decision Support

S1–S4 Team Roles
S1 — Product / Business Analyst
S2 — Backend / System Engineer
S3 — Frontend / UX Engineer
S4 — AI / Data / QA Engineer

Stakeholder

Stakeholder	Role	Main Need
Procurement Manager	Manages procurement operations	Visibility, cost control, supplier performance
Buyer	Handles purchasing activities	Faster RFQs, quotations, and POs
Category Manager	Manages procurement categories	Spend analysis and strategic sourcing
Supply Chain Manager	Oversees supply chain	Risk and supply visibility
Supplier Manager	Manages supplier relationships	Supplier performance and risk monitoring
Production Planner	Plans manufacturing	Material availability and production impact
Finance Manager	Controls procurement spending	Budget, cost, and payment visibility
Supplier	Provides goods/services	Orders, requirements, communication, status
Operations Manager	Oversees business operations	Overall operational visibility
Administrator	Manages the platform	Users, roles, integrations, configuration

System Scope

Procurement

- Purchase requisitions
- RFQ management
- Supplier quotation comparison
- Supplier selection
- Purchase order management

Supplier Management

- Supplier profiles
- Supplier onboarding
- Supplier performance
- Supplier scorecards
- Supplier risk scoring

Supply Risk

- Delivery-delay detection
- Supplier risk monitoring
- Material shortage alerts
- Risk prioritization

Inventory/Material Visibility

- Current stock
- Incoming materials
- Safety stock
- Material shortages
- Procurement-to-material impact

Decision Support

- Procurement dashboard
- Risk dashboard
- Alerts and notifications
- AI-assisted recommendations
- Alternative supplier comparison

Governance

- Role-based access
- Audit trail

- Data validation
- Basic security controls

Business Process Modeling

Supplier → Procurement → Inventory → Warehouse/Store → Production Impact → Risk Detection → Decision → Action

AS-IS PROCESS

Material Requirement
↓
Production / Store identifies need
↓
Inventory checked manually
↓
Procurement contacted
↓
RFQ sent to suppliers
↓
Supplier quotations received
↓
Quotations compared manually
↓
Supplier selected
↓
PO created
↓
Supplier confirms delivery
↓
Warehouse receives material
↓
Inventory updated
↓
Production / Store checks availability
↓
Problem discovered?

YES → Manual follow-up
NO → Continue

Manual follow-up
↓
Procurement / Inventory / Production coordinate separately

A supply-chain issue becomes visible to the relevant team only after significant manual coordination or after the problem has already affected inventory or operations.

TO-BE PROCESS

Material Requirement
↓
Inventory Check
↓
Procurement Request
↓
RFQ / Supplier Quotations
↓
Supplier Comparison
↓
Supplier Evaluation
↓
PO Created
↓
Supplier Confirmation
↓
Continuous Monitoring
↓

↓	↓	↓
Delivery Data	Inventory Data	Quality Data

↓
Risk Engine
↓
Risk Detected?
/ \
NO YES
↓ ↓
Monitor Impact Analysis
↓
Material Impact
↓
Production Impact
↓
Alert Generated
↓

AI Recommendation

↓

Human Decision

↓

Action

↓

Verification

↓

Audit Trail